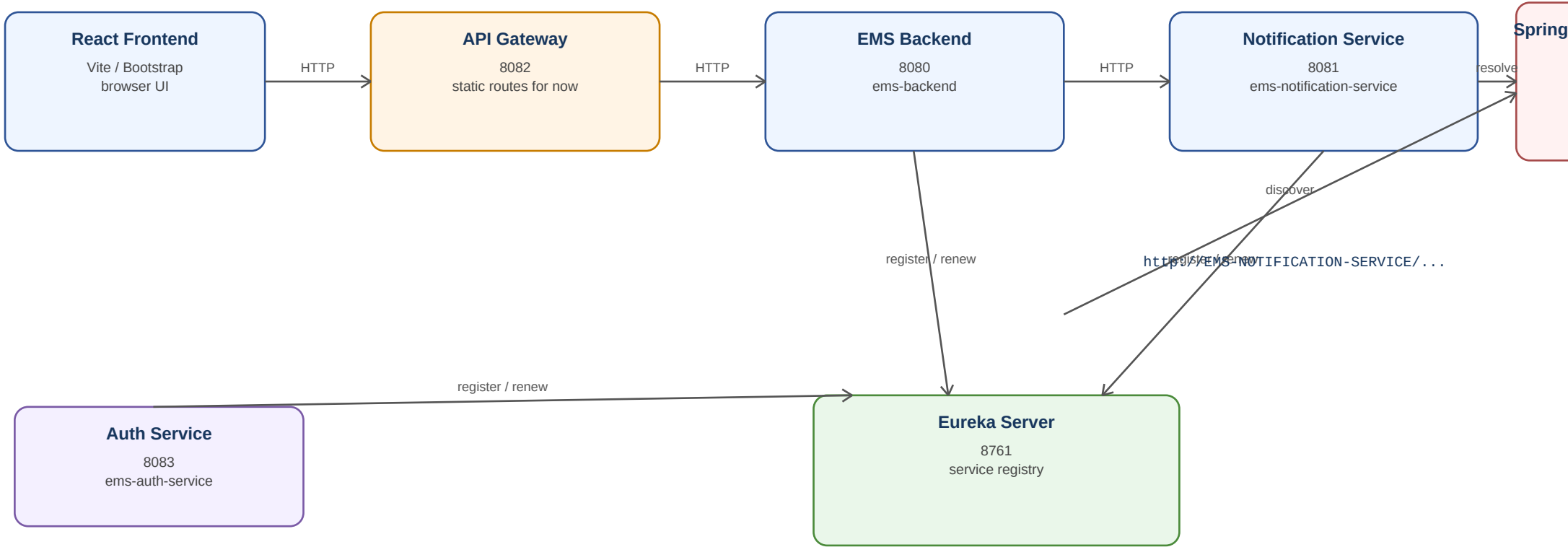


EMS — Eureka Service Discovery Block Diagram

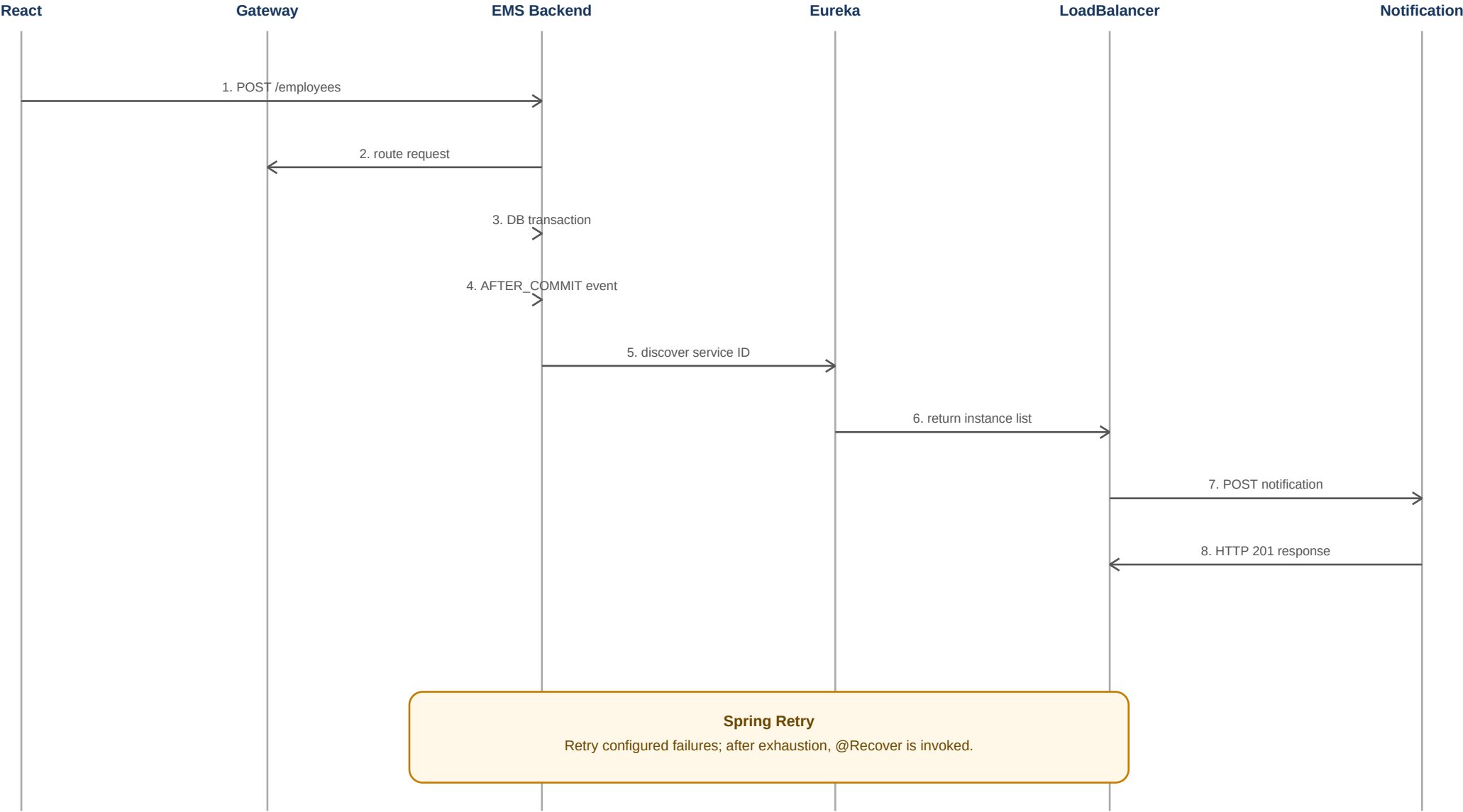
Current milestone: service registration + backend-side discovery + client-side load balancing



Note: Gateway is intentionally not Eureka-based yet. The next milestone is lb:// routing from Gateway.

Runtime Flow — Employee Creation to Notification

Discovery locates the service; LoadBalancer selects an instance; Spring Retry handles configured transient failures.



Why Eureka? — Before vs. After

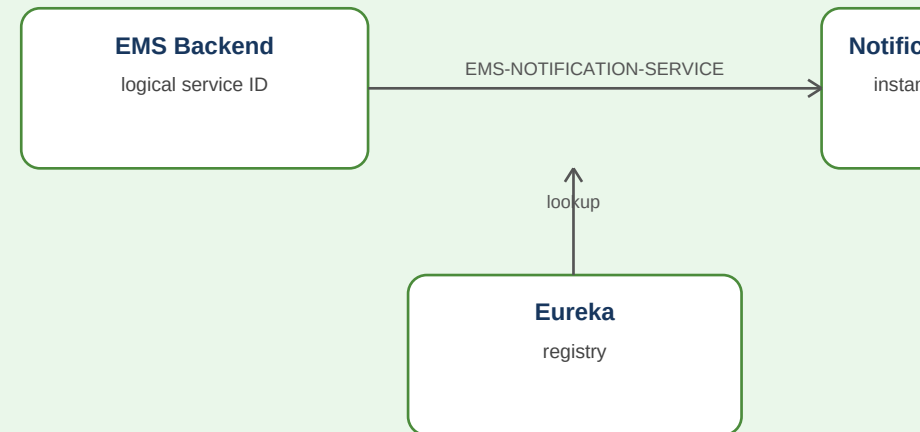
The architectural change is from physical endpoint dependency to logical service identity.

Before — fixed endpoint



- Consumer knows host + port.
- One fixed address represents one instance.
- Scaling or relocation requires changes.
- No registry-based lookup.

After — service discovery



- Services register themselves.
- Consumer uses logical service identity.
- LoadBalancer selects an instance.
- New instances can be added transparently.